

Student Enrollment & Course Management System (SECMS)(NM)

Student Enrollment & Course Management System (SECMS)

User Story:

The Student Enrollment & Course Management System (SECMS) is a Salesforce-based academic management solution developed to simplify and centralize the complete student enrolment lifecycle within an educational institution. The system provides a unified platform for managing student information, academic programs, course availability, instructor details, enrollment requests, approvals, and payment-related activities.

By utilizing Salesforce capabilities such as custom objects, relationships, automation, validation rules, approval processes, security controls, reports, and dashboards, SECMS reduces manual administrative work and improves the accuracy and accessibility of academic information. Administrators can maintain centralized student and course records, while enrollment officers can process enrollment requests and monitor student registrations efficiently. Instructors can access relevant course and student information to support academic activities.

The system also provides automated workflows for enrollment approvals, structured tracking of payments and course registrations, and real-time reports for monitoring institutional activities. Through a centralized Salesforce environment, SECMS improves coordination between students, enrollment staff, instructors, and administrators, creating a more organized, transparent, and efficient approach to academic enrollment management.

1.INTRODUCTION

1.1 Project Overview :

This project focuses on developing a centralized Student Enrollment & Course Management System using Salesforce to simplify and organize the academic administration activities of an educational institution. The application brings student information, course details, enrollment records, instructor assignments, payments, and approval activities together within a single platform, reducing data duplication and improving accessibility.

The system enables Enrollment Officers to efficiently handle student admissions, course registration, enrollment updates, and communication activities. Instructors can view their assigned courses and students, while administrators can monitor overall enrollment activities, academic operations, and pending requests. The application also supports automated instructor allocation, task generation, data validation, and structured **approval workflows to ensure that enrollment activities are processed consistently.**

Salesforce features such as Flows, Approval Processes, Validation Rules, Reports, and Dashboards are utilized to automate repetitive operations and maintain data accuracy. The system provides continuous visibility into important indicators such as total enrollments, student status, course demand, revenue, instructor allocation, and pending approvals. By integrating these functions into one Salesforce-based solution, the project provides a structured, transparent, and efficient approach to managing the complete student enrollment lifecycle.

1.2 Purpose :

The primary objective of the Student Enrollment & Course Management System (SECMS) is to transform traditional academic enrollment activities into a centralized, automated, and secure Salesforce-based solution. The system is designed to improve the overall management of student and course information while providing a consistent workflow for enrollment-related operations. The key objectives include:

- Centralized Student Information – Maintain complete and organized student records in a single Salesforce environment.
- Efficient Enrollment Management – Provide a systematic workflow for handling course registrations, enrollment requests, and academic allocations.
- Automated Processing and Validation – Reduce manual effort through automated approvals, validation rules, task creation, and workflow processes.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Enrollment Officer	manage student enrollments efficiently	enrollment data is scattered	systems are manual and unintegrated	frustrated and error-prone
PS-2	Administrator	track enrollments and revenue	reporting is delayed	data is not centralized	uncertain in decision-making



2. IDEATION PHASE

2.1 Problem Statement – Customer Problem Statement Template

Student Enrollment & Course Management System (SECMS)

PS-1 | Enrollment Officer



- I am (Customer)** : Enrollment Officer
- I'm trying to** : Process student admissions and course registrations efficiently.
- But** : Enrollment information is scattered.
- Because** : Manual and repetitive processes.
- Which makes me feel** : Stressed and concerned about errors.

PS-2 | Administrator



- I am (Customer)** : Administrator
- I'm trying to** : Monitor enrollments, payments and course capacity.
- But** : Information is difficult to track.
- Because** : Reports and records are distributed.
- Which makes me feel** : Uncertain about timely decisions.

PS-3 | Instructor



- I am (Customer)** : Instructor
- I'm trying to** : Manage assigned courses and enrolled students.
- But** : Student/course details are not readily available.
- Because** : Instructor allocation is handled separately.
- Which makes me feel** : Delayed in academic preparation.

PS-4 | Student



- I am (Customer)** : Student
- I'm trying to** : Register for courses and track approval status.
- But** : Enrollment progress is unclear.
- Because** : Communication is disconnected.
- Which makes me feel** : Uncertain about registration status.

2.2 Empathy Map Canvas

User Persona: Enrollment Officer

- **Think & Feel:** Concerned about double bookings and revenue loss
- **Hear:** Complaints from students and management
- **See:** Multiple spreadsheets and emails
- **Say & Do:** Rechecks data, follows up manually
- **Pain:** Duplicate entries, approval bottlenecks
- **Gain:** Automation, dashboards, faster processing



User Persona:
Enrollment Officer

"I just want the enrollment process to be smooth, accurate and stress-free."

Think & Feel

- Concerned about double bookings and revenue loss.

Hear

- Complaints from students and management.

See

- Multiple spreadsheets and emails.

Say & Do

- Rechecks data, follows up manually.

Pain

- Duplicate entries, approval bottlenecks.

Gain

- Automation, dashboards, faster processing.

2.3 Brainstorming & Idea Prioritization

Step 1 – Brainstormed Ideas

- Spreadsheet automation

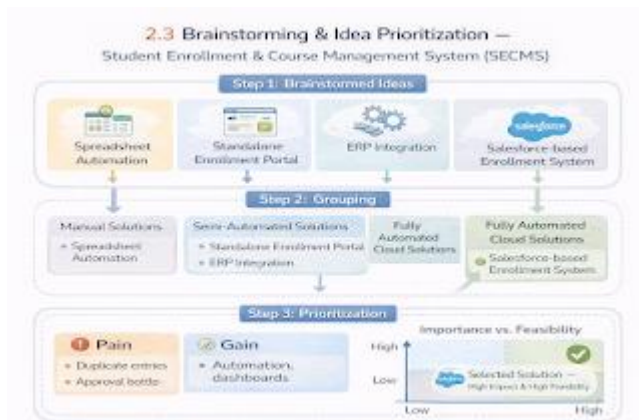
- Standalone enrollment portal
- ERP integration
- Salesforce-based enrollment system

Step 2 – Grouping

- Manual solutions
- Semi-automated solutions
- Fully automated cloud solutions

Step3–Prioritization

Salesforce-based SECMS was selected due to **high feasibility** and **high impact**.



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages covered:

- Awareness & Registration
- Course Selection
- Enrollment Request
- Fee Payment
- Approval
- Instructor Assignment
- Reporting & Completion



3.2 Solution Requirements

Functional Requirements

FR No	Epic	Description
FR-1	Student Management	Manage student records
FR-2	Course Management	Create and manage courses
FR-3	Enrollment Processing	Request, approve, reject enrollments
FR-4	Instructor Assignment	Auto-assign instructors
FR-5	Reporting	Dashboards and reports

Non-Functional Requirements

NFR-1: Usability

The system shall provide a simple, intuitive, and user-friendly interface that enables enrollment officers, administrators, instructors, and students to perform operations efficiently with minimal training. Navigation shall be organized through clearly labeled tabs, dashboards, and forms to reduce user effort and errors.

Examples:

- Quick student registration form
- One-click enrollment approval
- Role-based dashboards

NFR-2: Security

The system shall ensure the confidentiality, integrity, and protection of student and institutional data through secure authentication, authorization, and access controls. Sensitive information such as personal data, enrollment history, and academic records shall only be accessible to authorized users.

Examples:

- Only admins can modify course details
- Instructors view only assigned students
- Students cannot access other student records

NFR-3: Reliability

The system shall operate consistently and accurately without failures during enrollment processing, approvals, and reporting. Automation logic shall execute correctly to avoid duplicate or incorrect enrollments.

Examples:

- No duplicate course registration
- Accurate status updates after approvals
- Consistent report generation

NFR-4: Performance

The system shall provide fast response times for all operations to ensure smooth and uninterrupted user experience, especially during peak admission periods.

Examples:

- Quick dashboard refresh
- Fast search for student records
- No lag during bulk uploads

NFR-5: Scalability

The system shall support future growth in the number of students, courses, departments, and users without degradation in performance or functionality. It shall allow easy addition of new modules and features.

Examples:

- Add new departments without redesign
- Handle increased admissions annually
- Extend to mobile or portal integration

NFR-6: Availability

The system shall be continuously accessible to users during institutional working hours and must minimize downtime. The platform shall support high availability for academic operations.

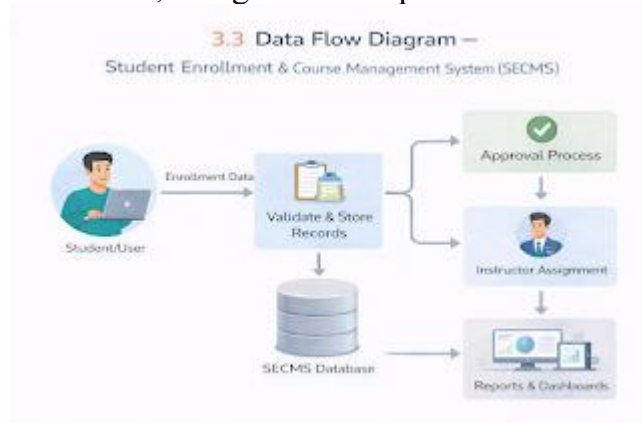
Examples:

- 24/7 cloud access
- Reliable during peak admission season
- Automatic backup of student records

3.3 Data Flow Diagram

- User inputs student and enrollment data
- System validates and stores records
- Approval process routes data
- Reports and dashboards consume processed data.

The system processes enrollment requests, validates data, routes approvals, assigns instructors, and generates reports.



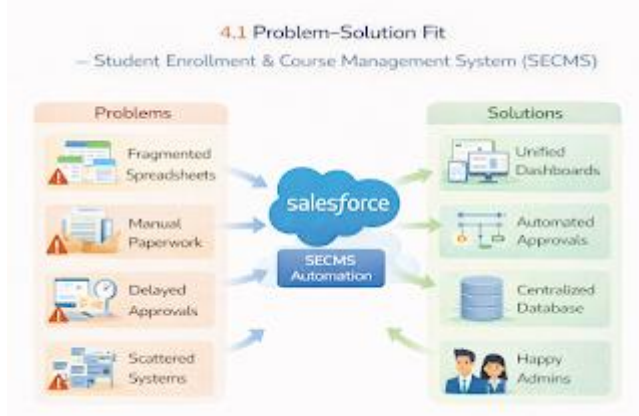
3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Approval Process
Reporting	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

SECMS directly addresses fragmented enrollment workflows by aligning user behavior, constraints, and automation using Salesforce-native capabilities.



4.2 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Manual enrollment handling
Solution	Salesforce-based SECMS
Novelty	End-to-end automation
Social Impact	Improved student experience
Revenue Model	Institutional deployment
Scalability	Multi-department support

4.3 Solution Architecture

The architecture integrates Salesforce objects, automation flows, approval processes, dashboards, and security layers.



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create a Salesforce developer account so that I can configure and deploy the Student Enrollment system..	1	High	Member1
Sprint-1	Data Modeling	USN-2	As an admin, I want to create custom objects for Student, Course, Enrollment, and Instructor so that academic data can be stored and managed efficiently.	3	High	Member2
Sprint-2	Data Modeling	USN-3	As a user, I want dedicated tabs for each object so that I can easily navigate between students, courses, enrollments, and instructors.	3	High	Member3
Sprint-2	Data Modeling	USN-4	As an admin, I want to create relevant fields such as student details, course information, and enrollment status so that complete records can be captured accurately.	5	High	Member4
Sprint-2	Data Modeling	USN-5	As an enrollment officer, I want a centralized Lightning App so that I can access all academic operations from one interface.	3	High	Member4
Sprint-2	Data Modeling	USN-6	As an admin, I want to define relationships between students, courses, and enrollments so that the system can logically link and track academic activities.	5	Medium	Member1

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2	Data Modeling	USN-7	As a user, I want customized page layouts so that only relevant information is displayed, improving clarity and ease of data entry.	5	High	Member2
Sprint-3	Automation	USN-8	As a system, I want to validate input data such as mandatory fields and enrollment limits so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-9	As a system, I want automated flows to handle enrollment processing and notifications so that manual effort is reduced and processes become faster.	3	High	Member4
Sprint-3	Automation	USN-10	As an administrator, I want an approval process for enrollment requests so that registrations are verified before confirmation.	5	High	Member1
Sprint-4	Security	USN-11	As an admin, I want to configure role-based security so that users can only access data relevant to their responsibilities.	5	High	Member2
Sprint-5	Reports	USN-12	As a manager, I want reports on student enrollments, course registrations, and instructor assignments so that I can monitor academic operations.	4	High	Member3
Sprint-6	Dashboards	USN-13	As a manager, I want visual dashboards showing enrollment statistics and trends so that I can quickly analyze system performance.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase:

Project Flow:

- Milestone 1: Salesforce Developer Account Creation
- Milestone 2: Custom Object Creation (Student, Course, Enrollment, Instructor)
- Milestone 3: Tabs Creation
- Milestone 4: Lightning App Creation
- Milestone 5: Fields Creation
- Milestone 6: Create Relationships
- Milestone 7: Page Layout Customization
- Milestone 8: Validation Rules
- Milestone 9: Flows (Automation)
- Milestone 10: Approval Process
- Milestone 11: Reports
- Milestone 12: Dashboards
- Milestone 13: Security Setup
- Milestone 14: Conclusion

What you'll learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Creating approval workflows
7. Developing reports & dashboards
8. Setting up security with profiles, permission sets, and sharing
9. End-to-end application testing

Milestone 1-Salesforce :

Introduction:

Are you new to Salesforce? Not sure exactly what it is, or how to use it? Don't know where you should start on your learning journey? If you've

answered yes to any of these questions, then you're in the right place. This module is for you.

Welcome to Salesforce! Salesforce is game-changing technology, with a host of productivity-boosting features, that will help you sell smarter and faster. As you work toward your badge for this module, we'll take you through these features and answer the question, "What is Salesforce, anyway?"

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud. So what does that really mean? Well, before Salesforce, your contacts, emails, follow-up tasks, and prospective deals might have been organized something like this:

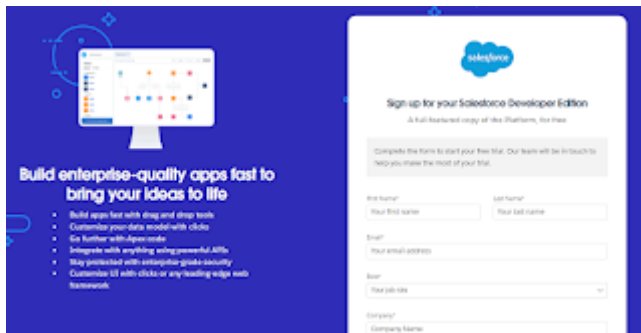
<https://youtu.be/r9EX3IGde5k>

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

Go to <https://developer.salesforce.com/signup>

On the sign up form, enter the following details :



First name & Last name

Email

Role : Developer

Company : College Name

County : India

Postal Code : pin code

Username : should be a combination of your name and company

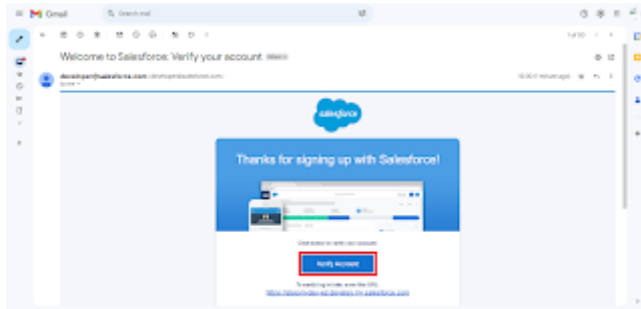
This need not be an actual email id, you can give anything in the format :

username@organization.com

Click on sign me up after filling these.

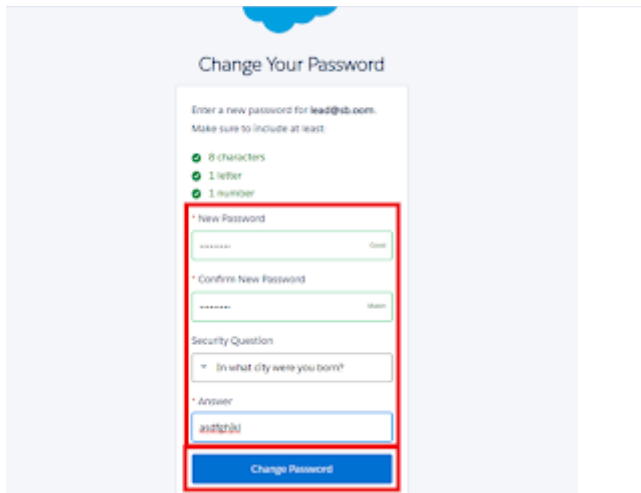
Activity 2: Account Activation:

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.

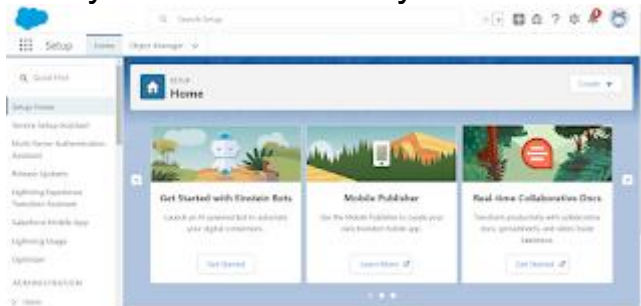


2. Click on Verify Account

3. Give a password and answer a security question and click on change password.



4. Then you will redirect to your salesforce setup page.



Milestone 2 – Objects:

Activity 1: Creating Student Object:

The purpose of creating the Student custom object is to store and manage information about students such as their personal details, contact information, and enrollment status.

To create the Student object:

Go to the Setup page.

Click on Object Manager.

Click on Create → Custom Object.

Enter the details as below:

1. Label: Student
2. Plural Label: Students
3. Object Name: Student
4. Record Name: Student Name
5. Data Type: Text
6. Under Optional Features, select:
7. Allow Reports
8. Allow Activities (optional but recommended)
9. Track Field History (optional, if you want)
10. Under Search Status, select:
11. Allow Search
12. Click Save.

The image shows two screenshots of the Canvas LMS 'Custom Object Definition' setup page for 'Student'.

Top Screenshot: Custom Object Definition Label

- Custom Object Information:**
 - Record Name:** Student Name (highlighted with a red box)
 - Record Name:** Student Name (highlighted with a red box)
 - Record Name:** Student Name (highlighted with a red box)

Bottom Screenshot: Optional Features and Search Status

- Optional Features:**
 - ☒ Allow Reports (highlighted with a red box)
 - ☒ Allow Activities (highlighted with a red box)
 - ☒ Track Field History (highlighted with a red box)
- Search Status:**
 - ☒ Allow Search (highlighted with a red box)

At the bottom of the page, there are buttons for 'Save', 'Save & New', and 'Cancel'.

In the same way Create **Courses**, **Instructors**, and **Enrolments** objects

Milestone 3 - Fields:

Table 1: Student Object Fields:

Field Label	Data Type
Student Name	Text (Standard)
Email	Email

Phone	Phone
Date of Birth	Date
Student Status	Picklist
Address	Text Area

Table 2: Course Object Fields:

Field Label	Data Type
Course Name	Text (Standard)
Category	Picklist
Duration	Number/Text
Course Fee	Currency
Description	Text Area

Table 3: Instructor Object Fields:

Field Label	Data Type
Instructor Name	Text (Standard)
Instructor Code	Text
Expertise	Picklist
Phone	Phone
Email	Email

Table 4: Enrollment Object Fields:

Field Label	Data Type
Student	Lookup (Student)
Course	Lookup (Course)
Instructor	Lookup (Instructor)
Enrollment Date	Date
Enrollment Status	Picklist
Fees Paid	Checkbox
Total Amount	Currency
Comments	Text Area

Milestone 4 – Create Relationships:

1. Activity 1: Create Lookup from Enrollment to Student:

1. Go to Setup → click Object Manager.
2. Type Enrollment in Quick Find → click Enrollment__c object.
3. Click Fields & Relationships → click New.
4. Select Lookup Relationship → click Next.
5. For Related To, select Student__c → click Next.
6. Enter Field Label: Student (API Name: Student__c).
7. Click Next through Field-Level Security and Layout options (ensure it's visible and added to layout).
8. Click Save.

Activity 2: Create Lookup from Enrollment to Course

Purpose:

To link each Enrollment to a Course.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Fields & Relationships → New.
3. Select Lookup Relationship → Next.
4. Related To: select Course__c → click Next.
5. Field Label: Course (API Name: Course__c).
6. Click Next (set visibility and layout) → Save.

Activity 3: Create Lookup from Enrollment to Instructor:

Purpose:

To store which Instructor is assigned to a particular Enrollment.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Fields & Relationships → New.
3. Choose Lookup Relationship → Next.
4. Related To: select Instructor__c → Next.
5. Field Label: Instructor (API Name: Instructor__c).
6. Complete Field-Level Security and Page Layout options → Save

Activity 2: Fees Must Be Paid Before Approval

Purpose:

Prevent Enrollment from being approved until fees are paid.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Validation Rules → New.
3. Rule Name: Fees_Must_Be_Paid_Before_Approval.
4. Error Condition Formula:

```
AND(
    ISPICKVAL(Enrollment__Status__c, "Approved"),
    Fees_Paid__c = FALSE
)
```

5. Error Message: Fees must be paid before approving an enrollment.

6. Error Location: Top of Page (or field as needed).

Click Save.

Milestone 6 – Approval Process

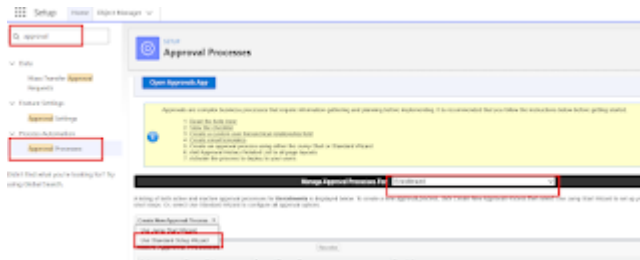
Activity 1: Create Approval Process for Enrollment Requests

Purpose:

Route Enrollment requests to the Training Manager for approval or rejection.

Steps:

1. Go to Setup → Approval Processes → select Enrollment__c.
2. Click Create New Approval Process → Use Standard Setup.
3. Process Name: Approval_Enrollment_Request.
4. Entry Criteria: Enrollment__Status__c = "Requested".
5. Save and configure steps.



Approval Process Edit Approval Enrollment Request

Step 1. Enter Name and Description

Enter a name and description for your new approval process.

Enter Name and Description

Process Name	Approval Enrollment Request
Unique Name	Approval_Enrollment_Request
Description	Approval process for Enrollment requests — sends to Training Manager

Step 2. Specify Entry Criteria

If only certain types of records should enter this approval process, enter that criteria below. For example, only requests from employees at headquarters should use this approval process.

Specify Entry Criteria

Use this approval process if the following criteria are met:

Field	Operator	Value
Enrollment: Enrollment Status	Is Equal To	Pending
Name	Is Equal To	Training Manager
Enrollment: Enrollment Status	Is Not Equal To	Rejected
Name	Is Not Equal To	Training Manager

6. Add Approval Step:
7. Approver: specific Training Manager user.

Approval Processes

Approval Process Edit
Training Manager

Step 1. Enter Name and Description

Enter a name, description, and class number for your new approval process.

Enter Name and Description

Name	Approval Enrollment Request
Unique Name	Approval_Enrollment_Request
Description	Approval process for Enrollment requests — sends to Training Manager

8. Under Final Approval Actions:
9. Add Field Update to set Enrollment_Status__c = "Approved".

Field Updates

Define the field update, including the object associated with the workflow rule, approval process, or enrollment process, the field to update, and the value to apply. Note that the fields are shown only for the type that you select.

Field Update (Edit)

Save Save & New Cancel

Identification

Name	Status Is Approved
Unique Name	Status_Is_Approved
Description	

Specify New Field Value

Field Update

Field to Update: Enrollment: Enrollment Status

Field Data Type: Picklist

Do not evaluate Workflow Rules after Field Change: ☐

Specify New Field Value

Field Update Options

☐ The value above the current one

☐ The value below the current one

☒ A specific value: Approved

Save Save & New Cancel

10. Under Final Rejection Actions:
11. Add Field Update to set Enrollment_Status__c = "Rejected".

12. Add Email Alert to send rejection email to student.

13. Activate the approval process.

Milestone 7- Tabs

Activity 1: Creating a Tab for the Student Object:

Go to Setup.

1. In the Quick Find box, type Tabs.
2. Under Custom Object Tabs, click New.
3. Select Student from the object list.
4. Choose a tab style (icon).
5. Click Next, assign it to all profiles, and then click Save.

Activity 2: Creating Remaining Tabs

Now create the Tabs for the remaining Objects, they are “Courses, Instructors, Enrollment”.

Follow the same steps as mentioned in Activity -1

Milestone 8 - The Lightning App

Activity 1: Creating the Lightning Application for the Project

Navigate to Setup.

Enter App Manager in the Quick Find search box.

Click New Lightning App.

Provide the app details:

App Name: Student Enrollment & Course Management

Developer Name auto-fills

Description: Optional, but can describe the purpose of the app

Click Next and choose the following settings:

App Branding (optional): Add logo or select color theme

Navigation Style: Choose Standard Navigation

Click Next and assign the app to specific profiles (e.g., **Admin, Enrollment Officer, Instructor**).

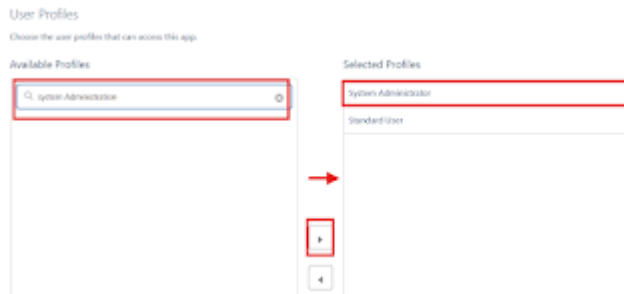
Click Next to select the items (tabs) to be included in the app. Add:

1. Student
2. Course
3. Instructor
4. Enrollment
5. Reports
6. Dashboards
7. Home
8. Tasks

Click Next and review the app settings.

Finally, click Save & Finish.

The screenshot displays the 'Lightning Experience App Manager' interface. The 'App Details & Branding' section is active, showing the app name 'Student Enrollment & Course Management' (highlighted with a red box), developer name 'Student_Enrollment_Course_Management', and a description field. The 'App Branding' section shows an image placeholder and a primary color hex value of '#0070C2'. The 'Navigation Items' section is also visible, showing a list of available items (Accounts, Contacts, Dashboards, etc.) and a 'Selected Items' list (Accounts, Contacts, Dashboards, etc.) (highlighted with a red box).



Milestone 9 – Page Layouts and Record:

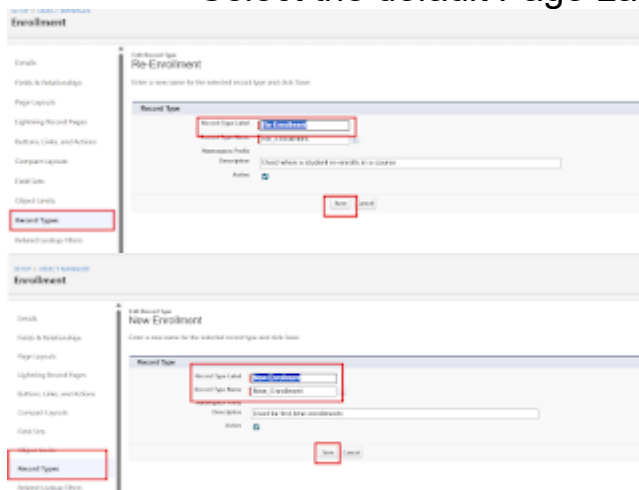
Activity 1: Create Record Types for Enrollment:

Purpose:

To separate New Enrollment vs Re-Enrollment processes.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Record Types → New.
3. For the first record type:
 - Existing Record Type: Master
 - Record Type Label: New Enrollment
 - Enable for required profiles (Admin, Enrollment Officer).
 - Select the default Page Layout → Save & New.
4. For the second record type:
 - Record Type Label: Re-Enrollment
 - Enable for the same profiles.
 - Select the default Page Layout → Save.



Activity 2: Separate Page Layout Creators for Each Enrollment Record Type

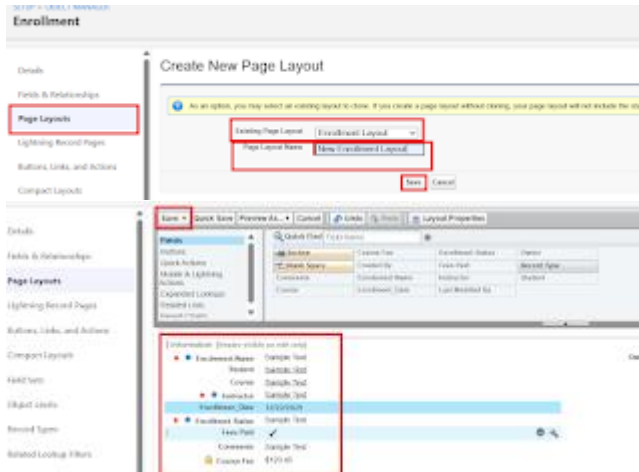
Purpose:

To show different fields for New Enrollment vs Re-Enrollment.

Steps:

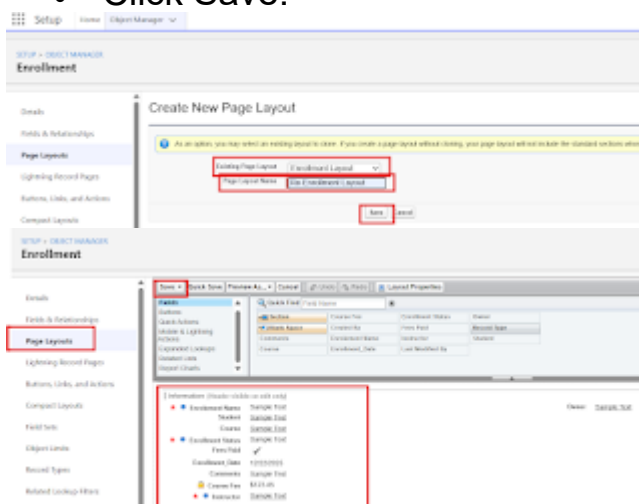
1. Enrollment__c object → click Page Layouts.
2. Click New to create a layout for New Enrollment:

- Clone from existing Enrollment Layout → name it New Enrollment Layout → Save.
- Keep fields relevant for first-time enrollment: Student, Course, Instructor, Enrollment Date, Status, Fees, etc.
- Remove unnecessary re-enrollment fields if any.
- Click Save.



3. Again, click New to create Re-Enrollment Layout:

- Clone existing layout → name it Re-Enrollment Layout → Save.
- Include fields like previous Enrollment reference (if any), Re-Enrollment Reason, Status, Fees, etc.
- Click Save.



4. Click Page Layout Assignment → Edit Assignment.

5. For each profile and record type:

- New Enrollment → use New Enrollment Layout.

6. Re-Enrollment → use Re-Enrollment Layout.

Click Save.

Activity 3: Enrollment Related Lists to Parent Objects:

Purpose:

To see all Enrollments from Student, Course, and Instructor records.

- a) Student → Enrollment Related List
Go to Setup → Object Manager → Student__c.
Click Page Layouts → open the main layout.
In the palette, select Related Lists.
Drag Enrollments related list onto the layout.
Customize columns if needed → Save.
- b) Course → Enrollment Related List
Go to Object Manager → Course__c → Page Layouts.
Open layout → add Enrollments related list.
Save.
- c) Instructor → Enrollment Related List
Go to Object Manager → Instructor__c → Page Layouts.
Open layout → add Enrollments related list.
Save.

Milestone 10 – Automation with Flow

Activity 1: Auto-Populate Enrollment Date (Record-Triggered Flow)

Purpose:

Set Enrollment_Date__c to today when Enrollment is created.

Steps:

1. Go to Setup → Flows → New Flow.
2. Select Record-Triggered Flow → Create.
3. Object: Enrollment__c.
4. Trigger: A record is created.
5. Optimize for Actions and Related Records → Done.

The screenshot shows the 'Select Object' and 'Configure Trigger' sections of the Salesforce Flow Builder. In the 'Select Object' section, 'Enrollment' is selected in the dropdown menu. In the 'Configure Trigger' section, the 'Trigger the Flow When:' options are shown, with 'A record is created' selected. Below this, the 'Set Entry Conditions' section is visible, showing 'Condition Requirements' set to 'None'.

Select Object

Select the object whose records trigger the flow when they're created, updated, or deleted.

* Object

Enrollment

Configure Trigger

Trigger the Flow When:

☒ A record is created

☐ A record is updated

☐ A record is created or updated

☐ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of records that trigger the flow to conserve your org's resources.

Condition Requirements

None

6. Add Update Records element:
 - Label: Set Enrollment Date.
 - Use a record that triggered the flow.

- Field: Enrollment_Date__c → Value: \$Flow.CurrentDate.

7. Save as SetEnrollmentDate → Activate.

The screenshot shows the 'Update Records' configuration screen. At the top, there's a 'Label' field with 'Set Enrollment Date' and an 'API Name' field with 'Set_Enrollment_Date'. Below is a 'Description' text area. A section titled '* How to Find Records to Update and Set Their Values' contains four radio buttons: 'Use the enrollment record that triggered the flow' (selected), 'Update records related to the enrollment record that triggered the flow', 'Use the IDs and all field values from a record or record collection', and 'Specify conditions to identify records, and set fields individually'. Below this is the 'Set Filter Conditions' section with a dropdown menu set to 'None—Always Update Record'. The 'Set Field Values for the Enrollment Record' section shows a table with two columns: 'Field' and 'Value'. The first row has 'Enrollment_Date' in the field column and 'Running Flow Interview > CurrentDate' in the value column.

Activity 2: Send Email When Enrollment Approved

Purpose:

When Enrollment_Status__c = Approved, send email, alert admin, and set Student to Active.

Steps (high level for document):

1. Create Email Template and Email Alert for student notification.
2. Go to Flows → New → Record-Triggered Flow.
3. Object: Enrollment__c, Trigger on update.
4. Condition: Enrollment_Status__c Equals "Approved".

The screenshot shows the 'Configure Start' screen. At the top, there's a 'Start' dropdown menu set to 'Enrollment'. Below is the 'Configure Trigger' section with four radio buttons: 'A record is created', 'A record is updated', 'A record is created or updated' (selected), and 'A record is deleted'. Below this is the 'Set Entry Conditions' section with a dropdown menu set to 'All Conditions Are Met (AND)'. At the bottom, there's a table with three columns: 'Field', 'Operator', and 'Value'. The first row has 'Enrollment_Status' in the field column, 'Equals' in the operator column, and 'Approved' in the value column.

5. Add Get Records to fetch related Student__c.

Get Records

*Label:

*API Name:

Description:

Get Records of This Object

*Data Source: ☒ Data Cloud Object ☐ Salesforce Object

*Object:

Filter Student Records

Condition Requirements:

Field	Operator	Value
Status	Equals	Active

+ Add Condition

6. Add Action (Email Alert) to send email to student.

Enrollment Approved EmailAlert

*Label:

*API Name:

Description:

Use values from earlier in the flow to set the inputs for the "Enrollment Approved EmailAlert" email alert. To use its outputs later in the flow, store them in variables.

Set Input Values

*Record ID:

7. Add Action to notify Admin if needed.

8. Add Update Records to set Student__c.Status__c = "Active".

Update Records

*Label:

*API Name:

Description:

*How to Find Records to Update and Set Their Values: ☒ Use the enrollment record that triggered the flow ☐ Update records related to the enrollment record that triggered the flow ☐ Use the IDs and all field values from a record or record collection ☐ Specify conditions to identify records, and set fields individually

Set Filter Conditions

Condition Requirements to Update Record:

Set Field Values for the Enrollment Record

Field	Value
Comments	Active

9. Save as Enrollment Approved Action → Activate.

Activity 3: Auto-Assign Instructor Based on Course Category

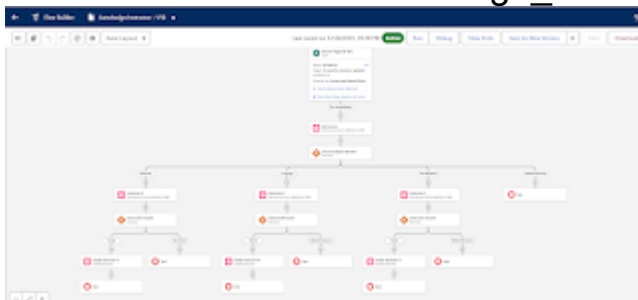
Purpose:

Assign Instructor automatically based on Course category.

Rules:

Technical → Instructor A
 Language → Instructor B
 Non-Technical → Instructor C
 Steps:

1. Prepare Instructor__c records with codes (e.g., INSTR_A, INSTR_B, INSTR_C).
2. Create Record-Triggered Flow on Enrollment__c (created/updated).
3. Add Get Records to fetch related Course__c.
4. Add Decision element on Course__r.Category__c:
5. Technical / Language / Non-Technical outcomes.
6. For each outcome:
7. Get Instructor record (A/B/C) based on code or name.
8. Update Records: set Instructor__c = Get_Instructor_X.Id.
9. Save as AutoAssign_Instructor → Activate.



Activity 4: Auto-Create Task for Follow-Up

Purpose:

When Enrollment is “Requested”, create a follow-up task for Admin.

Steps:

1. Create Record-Triggered Flow on Enrollment__c.
2. Trigger: Record created or updated; condition Enrollment_Status__c = "Requested".

Field	Operator	Value
Enrollment_Status__c	Equals	Requested

3. Add Get Records to fetch Admin user (by Username / Profile).
4. Add Formula (Date): TODAY() + 2 for due date.

5. Add Create Records for Task:
6. Subject: Follow up on enrollment request.
7. OwnerId: Admin user Id.
8. WhatId: Enrollment Id.
9. ActivityDate: formula Today + 2.

10. Save as Create_Followup_Task → Activate.

Milestone 11 – Reports:

Activity 1: Create “Students by Status” Report

1. Go to **Reports** → **New Report**.
2. Report Type: **Students**.
3. Group rows by **Status__c**.
4. Add columns as needed (Name, Email, Phone).
5. Save as **Students by Status**.

Name	Email	Phone
Student 1	student1@example.com	1234567890
Student 2	student2@example.com	9876543210
Student 3	student3@example.com	5555555555
Student 4	student4@example.com	1111111111
Student 5	student5@example.com	2222222222

Activity 2: Create “Enrollments by Course” Report

1. New Report → Report Type: **Enrollments**.
2. Group by **Course__r.Name**.
3. Show Record Count per course.
4. Save as **Enrollments by Course**



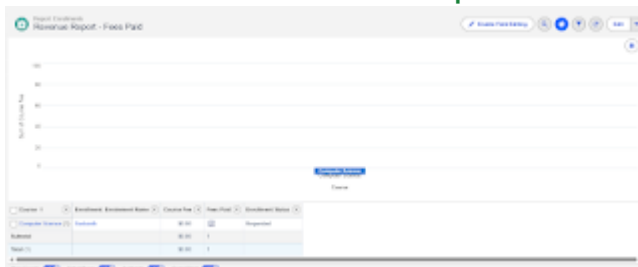
Activity 3: Create “Pending Enrollment Approvals” Report

1. New Report → **Enrollments** report type.
2. Filter: **Enrollment_Status__c = "Requested"**.
3. Add columns (Student, Course, Owner, Created Date).
4. Save as **Pending Enrollment Approvals**.



Activity 4: Create “Revenue Report (Fees Paid)” Report

1. New Report → **Enrollments** report type.
2. Filter: **Fees_Paid__c = TRUE**.
3. Group by **Course__r.Name**.
4. Summarize **Course_Fee__c** or similar fee field.
5. Save as **Revenue Report – Fees Paid**.



Milestone 12 – Dashboards:

Activity 1: Create “Student Management Dashboard”

Steps:

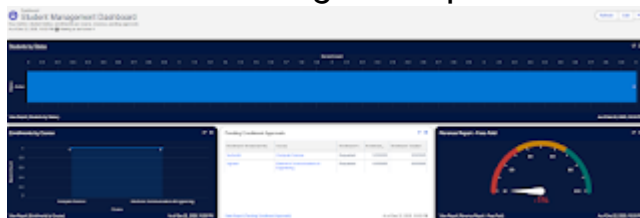
1. Go to **Dashboards** → **New Dashboard**.
2. Name: **Student Management Dashboard**.



3. Select folder → **Create**.
4. Select the new Dashboard folder and Save.

Add components:

- **Donut Chart** → using **Students by Status** report.
 - **Bar Chart** → using **Enrollments by Course** report.
 - **Gauge** → using **Revenue Report – Fees Paid** (sum of revenue field).
 - **Table** → using **Pending Enrollment Approvals** report.
4. Arrange components on canvas.



Click **Save** → **Done**.

Milestone 13 – Security Setup

Activity 1: Configure Profiles

):

1. Go to **Setup** → **Profiles** → **New** (or Clone)
Profiles Required:
 - **Admin** – Full access.
 - **Enrollment Officer** – Create/Update Enrollments.
 - **Instructor** – Read-only access to assigned Enrollments.
2. **Steps** (example for Enrollment Officer fName: **Enrollment Officer**).
3. In **Object Settings**:
 - **Enrollment__c**: Read, Create, Edit.
 - **Student__c** / **Course__c**: Read (and Edit if required).
4. Save.

Repeat similarly for **Instructor** profile with **Read-Only** on Enrollments

Activity 2: Create Permission Sets

1. Go to **Setup** → **Permission Sets** → **New**.
2. Create **Custom Object Access** (additional access to project objects if needed).
3. Create **Dashboard Editor** (Create/Customize Reports & Dashboards permissions).
4. Assign to relevant users.

Activity 3: Configure Sharing & OWD

Steps:

1. Go to **Setup** → **Sharing Settings**.
2. Set **Organization-Wide Defaults**:
 - **Students (Student__c)**: Public Read/Write.
 - **Courses (Course__c)**: Public Read/Write.
 - **Enrollments (Enrollment__c)**: Private.
3. Save.

For Instructor access to assigned Enrollments, optionally:

- Add a **User lookup** on Enrollment (Instructor User).
- Use a simple **Record-Triggered Flow** to set **OwnerId = Instructor User** so they can see only their own records.

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7. FUNCTIONAL & PERFORMANCE TESTING

The testing phase was conducted to verify that the **Student Enrollment & Course Management System (SECMS)** performs its intended functions accurately and consistently within the Salesforce environment. Each major module was tested through the Salesforce user interface to confirm proper record creation, automation, validation, and data processing.

Testing covered the creation and management of **Student, Course, Instructor, and Enrollment** records. Validation Rules were tested to ensure that mandatory information, eligibility requirements, enrollment limits, and other business conditions were properly enforced. Different **Record Types** were also verified to confirm that appropriate fields and processes were displayed according to the selected course category.

Salesforce **Flows** were tested for automated activities such as enrollment processing, seat availability verification, task generation, notifications, and approval-related operations. Where applicable, **Apex Triggers** were tested to verify automatic record updates, enrollment confirmation, and related business logic.

Reports and Dashboards were also reviewed to confirm that the displayed information accurately represents the underlying Salesforce records. Screenshots were captured throughout the testing process as evidence of successful feature execution and system validation.

Testing screenshots should be submitted for:

- Student record creation and management
- Course and instructor record creation
- Enrollment registration and updates
- Validation Rule execution
- Record Type behavior
- Flow automation and approval processing
- Seat availability verification
- Apex Trigger execution and automatic updates
- Reports and Dashboard results

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- **Centralized Information Management:** Student, course, instructor, enrollment, and payment-related information can be maintained within a unified Salesforce environment.
- **Workflow Automation:** Flows and approval processes reduce repetitive manual activities and help maintain consistent business operations.
- **Improved Data Accuracy:** Validation Rules help prevent incomplete, invalid, or inconsistent records.
- **Real-Time Reporting:** Reports and Dashboards provide updated information for monitoring enrollment activities and organizational performance.
- **Better Process Visibility:** Administrators and authorized users can track enrollment status, approvals, course allocation, and related activities more effectively.

8.2 Disadvantages

- **Salesforce Dependency:** The application depends on the Salesforce platform and its available features, limits, and service availability.
- **Initial Configuration Effort:** Designing objects, relationships, automation, validation rules, permissions, and dashboards requires considerable planning and configuration.
- **Platform-Based Costs:** Advanced Salesforce capabilities and increased usage may require suitable Salesforce licensing depending on organizational requirements.
- **Learning Requirement:** Administrators and users may require initial training to work effectively with the customized Salesforce application.

9. CONCLUSION

The **Salesforce Student Enrollment & Course Management System (SECMS)** provides a structured platform for managing major academic enrollment activities, including student registration, course management, instructor allocation, enrollment processing, approvals, revenue tracking, and reporting.

By combining **Salesforce Objects, Validation Rules, Flows, Approval Processes, Reports, and Dashboards**, the system reduces dependence on manual processes and provides better control over academic data. Automated workflows help standardize routine activities, while validation mechanisms improve data consistency and reporting tools provide meaningful visibility into enrollment operations.

Overall, the project demonstrates how Salesforce can be configured as a centralized and scalable solution for transforming traditional enrollment procedures into a more organized, automated, and transparent digital workflow.

10. FUTURE SCOPE

The system can be further enhanced with additional features to provide greater accessibility, automation, and analytical capabilities:

- **Online Payment Gateway:** Integration with secure payment services can enable students to complete course and enrollment payments directly through the system.
- **Student Self-Service Portal:** A dedicated portal can allow students to view courses, submit enrollment requests, track approvals, check payment status, and access relevant academic information.
- **Advanced Analytics:** AI and analytics capabilities can be introduced to identify enrollment patterns, course demand, revenue trends, and other useful operational insights.
- **Mobile Accessibility:** A mobile-friendly interface or Salesforce mobile implementation can provide users with access to important enrollment functions from smartphones and tablets.
- **Automated Notifications:** Email and in-app notifications can be expanded to provide timely updates regarding enrollment status, approvals, payments, and course changes.
- **External System Integration:** Future versions can integrate with learning management systems, accounting platforms, or other institutional applications.
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11. APPENDIX

11.1 Source Code and Configuration

The implementation consists primarily of Salesforce configuration components, including custom objects, fields, relationships, Record Types, Validation Rules, Flows, Approval Processes, Apex components, Reports, Dashboards, and user permissions.

11.2 Dataset

The system uses Salesforce records stored within the configured objects, including **Student, Course, Instructor, Enrollment, Payment, and related records** used for testing and reporting.

11.3 Project Demonstration

The completed system can be demonstrated through the configured Salesforce Developer Org, showcasing the implemented objects, automated workflows, validation mechanisms, approval processes, reports, and dashboards.

THANK YOU